Homework #2

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① For each part determine if the first vector given is a member of span(S)?

(a) $x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ and $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$ (1)

$$\text{span}(S) = a \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \quad (2)$$

So, to check if $x \in \text{span}(S)$, we need to solve the following system to see if we have valid a, b:

$$a \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad (3) \quad \Rightarrow$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad (4) \quad \Rightarrow$$

$$\text{rref} \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \end{array}\right] = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow$$

Yes, $x = 1 \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, $x \in \text{span}(S)$

(b) $x = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ $\quad S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$

Again similar method and expressions (1), (2), (3):

$$a \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \qquad (6)$$

Let's solve (6) to see $\exists\, a, b \in \mathbb{R}$ that satisfy (6):

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \qquad (7) \Rightarrow$$

$$\text{rref} \left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 1 & 2 \\ 2 & 1 & 1 \end{array}\right] = \text{rref} \left[\begin{array}{cc|c} 1 & -1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right] \qquad (8)$$

in (8), last row is implying $[0\ 0 \mid 1] \begin{bmatrix} a \\ b \end{bmatrix} = a \cdot 0 + b \cdot 0 = 1 \Rightarrow$ $0 = 1$ which is not valid. So, there is no $a, b \ni$ satisfy (6). $\Rightarrow$

No $x \notin \text{span}(S)$.

(c) $v = (-x^3 + 2x^2 + 3x + 3)$ $\quad S = \{(x^3 + x^2 + x + 1), (x^2 + x + 1), (x + 1)\}$

To check if $v \in \text{span}(S)$, we solve

$$a(x^3 + x^2 + x + 1) + b(x^2 + x + 1) + c(x + 1) = -x^3 + 2x^2 + 3x + 3 \Rightarrow$$

$$\begin{cases} a = -1 \\ b = 3 \\ c = 1 \end{cases} \Rightarrow \text{Yes, } v \in \text{span}(S)$$

(d) $v = 2x^3 - x^2 + x + 3$; $S = \{(x^3+x^2+x+1), (x^2+x+1), (x+1)\}$

To check if $v \in \text{span}(S)$, we solve the following system for a, b, c:

$$a(x^3+x^2+x+1) + b(x^2+x+1) + c(x+1) = 2x^3 - x^2 + x + 3 \qquad (9)$$

But (9) does not have any solutions for (a, b, c). Hence, No, $v \notin \text{span}(S)$

$\square$

② Show that the matrices

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

generate $M_{2\times2}(F)$

Solution:

Consider an arbitrary matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times2}(F)$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \underbrace{\begin{bmatrix} \boxed{a}+0 & \boxed{0}+b \\ \boxed{0}+c & \boxed{0}+d \end{bmatrix}}_{M_{2\times2}(F)} = \underbrace{\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}}_{\in M_{2\times2}(F)} + \underbrace{\begin{bmatrix} 0 & b \\ c & d \end{bmatrix}}_{\in M_{2\times2}(F)}$$

I'll apply same technique on second term

$$= \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} \boxed{0}+0 & \boxed{b}+0 \\ \boxed{0}+c & \boxed{0}+d \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ c & d \end{bmatrix} =$$

Scalar multiplication vector addition

$$a\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + \begin{bmatrix}b\cdot 0 & b\cdot 1\\ b\cdot 0 & b\cdot 0\end{bmatrix} + \begin{bmatrix}\boxed{0}+0 & \boxed{0}+0\\ \boxed{c}+0 & \boxed{0}+d\end{bmatrix} =$$

$$= a\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + b\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\ c & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\ 0 & d\end{bmatrix}$$

I'll use same scalar multip-n rule for last 2 terms

$$= a\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + b\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} + c\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix} + d\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix} \qquad (10)$$

Since (10) is true for any arbitrary matrix A in $M_{2\times 2}(F)$, the given 4 vectors generate $M_{2\times 2}(F)$.

$\square$

③ Show that a subset W of a vector space V is a subspace of V if and only if

$$\text{span}(W) = W.$$

Solution:

($\Longleftrightarrow$)

Let V be a vector space. (1) &

Assume a subset W is a subspace of V. (2)

We need to show (WTS): $\text{span}(W) = W$. $\Rightarrow$

I'll use subset inclusion: $\begin{cases}\text{span}(W) \subseteq W\\ W \subseteq \text{span}(W)\end{cases}$ (2')

For $W \subseteq span(W)$:

If we take an arbitrary vector $w \in W$, since W is a subspace, W is closed under addition & scalar multiplication.

If $w = \vec{0} \in W$, it is trivial that $w = \vec{0} \in span(W) \Rightarrow$

$$W \subseteq span(W) \qquad (3)$$

If $w \neq \vec{0} \in W$, because W is a subspace, $\exists$ vectors $u \in W$, $v \in W$ and scalars $a, b \in F$ such that

$$w = a \cdot u + b v \qquad (4) \Rightarrow$$

$w \in span(W)$.

Since (4) holds for any arbitrary vector $w \in W$, $\Rightarrow$ $w \in span(W) \Rightarrow$

$$W \subseteq span(W) \qquad (5)$$

Now, let me try reverse version.

For $\text{span}(W) \subseteq W$

Take any arbitrary vector $w \in \text{span}(W)$.

If $w = 0$, it is obvious $0 \in W$ since W is a subspace.

If $w \neq 0 \in \text{span}(W)$, $\exists$ vectors $w_1, w_2 \ldots w_n \in W$ & scalars $a_1, a_2 \ldots a_n \in F$ $\ni$

$$w = a_1 w_1 + a_2 w_2 + \ldots + a_n w_n \qquad (6)$$

Since W is a subspace, it is closed under addition & scalar multiplication. $\Rightarrow$

$$\underbrace{\left[\ldots\left(\left(\underbrace{\overbrace{a_1 w_1}^{\in W} + \overbrace{a_2 w_2}^{\in W}}_{\in W}\right) + \underbrace{a_3 w_3}_{\in W}\right) + \underbrace{a_4 w_4}_{\in W}\right) + \ldots + a_{n-1} w_{n-1}\right]}_{\in W} + \underbrace{a_n w_n}_{\in W} \in W$$

$\Rightarrow \forall w \in \text{span}(W) \Rightarrow w \in W$ which means $\text{span}(W) \subseteq W$ (7)

By (5) & (7), we conclude that $\text{span}(W) = W$.

($\Leftarrow$)

Now, we try reverse direction for if and only if.

Let V be a vector space and W be a subset of V.

Assume $\text{span}(W) = W$. (8)

WTS : W is a subspace.

Theorem 1.3 (For a subspace)
Let V be a vector space and W a subset of V. Then W is a subspace of V if and only if the following 3 conditions hold for the operations defined in V.

(i) $0 \in W$.

(ii) $(x+y) \in W \quad \forall x, y \in W$.

(iii) $(cx) \in W \quad \forall c \in F, \forall x \in W$.

Using this theorem 1.3, I am going to check if W passes these (i) - (iii) tests:

(i) $0 \in W$ since $0 \in \text{span}(W) = W$ (9)

(ii) $\forall x, y \in W$, $\exists$ vectors $x_1, x_2, \ldots x_n$, $y_1, y_2, \ldots y_m$ & scalars $a_1, \ldots a_n$, $b_1, b_2, \ldots b_m$ $\ni$

$$+\begin{cases} x = a_1 x_1 + a_2 x_2 + \ldots + a_n x_n \in \text{span}(W) \\ y = b_1 y_1 + b_2 y_2 + \ldots + b_m y_m \in \text{span}(W) \end{cases} \Rightarrow$$

$$x + y = \left[\underbrace{a_1 x_1 + \ldots + a_n x_n}_{\in \text{span}(W)} + \underbrace{b_1 y_1 + \ldots + b_m y_m}_{\in \text{span}(W)}\right] \in \text{span}(W) = W \Rightarrow$$

$(x+y) \in W$ (10)

(iii) Finally $\forall c \in F$, and the vector x above:

$$cx = c(a_1 x_1 + a_2 x_2 + \ldots + a_n x_n) \overset{(VS7) \,\&\, (VS6)}{=\!=\!=}$$

$$= (ca_1)x_1 + (ca_2)x_2 + \ldots + (ca_n)x_n \in \text{span}(W) = W \Rightarrow$$

$(cx) \in W$ (11)

(9), (10), (11) tell us W is a subspace.

$\square$

④ Let S_1, S_2 be subsets of a vector space V. Prove that

$$\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2) \quad (1)$$

Give an example in which

$$\text{span}(S_1 \cap S_2) \neq \text{span}(S_1) \cap \text{span}(S_2) \quad \&$$

$$\text{span}(S_1 \cap S_2) = \text{span}(S_1) \cap \text{span}(S_2)$$

Solution:

Let S_1, S_2 be subsets of a vector space V. Then

WTS: $\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2) \quad (1)$

I am going to use subset inclusion method to prove (1):

$$\forall\, x \in \text{span}(S_1 \cap S_2) \quad (2)$$

Then $\exists$ vectors $u_1, u_2, \ldots, u_n \in (S_1 \cap S_2)$ (3)
and scalars $a_1, a_2, \ldots, a_n \in F$ $\ni$

$$\underbrace{x = a_1 u_1 + a_2 u_2 + \ldots + a_n u_n}_{\text{due to "definition of span}(S_1 \cap S_2)\text{"}} \quad (4)$$

④ Let S_1, S_2 be subsets of a vector space V. Prove that

$$\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2) \quad (1)$$

Give an example in which

$$\text{span}(S_1 \cap S_2) \neq \text{span}(S_1) \cap \text{span}(S_2) \quad \&$$

$$\text{span}(S_1 \cap S_2) = \text{span}(S_1) \cap \text{span}(S_2)$$

Solution:

Let S_1, S_2 be subsets of a vector space V. Then

WTS: $\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2) \quad (1)$

I am going to use subset inclusion method to prove (1):

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Then $\exists$ vectors $u_1, u_2, \ldots, u_n \in (S_1 \cap S_2)$ (3)
and scalars $a_1, a_2, \ldots, a_n \in F$ $\ni$

$$\underbrace{x = a_1 u_1 + a_2 u_2 + \ldots + a_n u_n}_{\text{due to "definition of span}(S_1 \cap S_2)\text{"}} \quad (4)$$

Since $\{u_1, u_2 \dots u_n\} \in (S_1 \cap S_2) \Rightarrow$

"definition of intersection."
$$\begin{cases} \{u_1, u_2 \dots u_n\} \in S_1 & (5) \quad \& \\ \{u_1\ u_2 \dots u_n\} \in S_2 & (6) \end{cases}$$

Considering (5), for coefficients $a_1 \dots a_n$, we can write

$$x = (a_1 u_1 + a_2 u_2 + \dots + a_n u_n) \in \text{span}(S_1) \quad (7)$$

And

Considering (6) and coefficients $a_1, a_2 \dots a_n \in F$, we can write

$$x = (a_1 u_1 + a_2 u_2 + \dots + a_n u_n) \in \text{span}(S_2) \quad (8)$$

(7) and (8) imply $x \in \text{span}(S_1) \cap \text{span}(S_2)$ (9)
because of the definition of intersection.

Because x, an arbitrary vector in $\text{span}(S_1 \cap S_2)$, belongs to $\text{span}(S_1) \cap \text{span}(S_2)$, it follows that

$$\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2) \quad (10).$$

Examples:

a) $\text{span}(S_1 \cap S_2) \neq \text{span}(S_1) \cap \text{span}(S_2)$ (11)

Let $S_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & $S_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ $\Rightarrow$ (12)

$S_1 \cap S_2 = \emptyset \Rightarrow \text{span}(\emptyset) = \{0\}$ (13)

Yet

$\text{span}(S_1) \cap \text{span}(S_2) = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) \cap \text{span}\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = c\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \forall c \in \mathbb{R}$ (14)

But, by (13) & (14) $\Rightarrow$

$\text{span}(S_1 \cap S_2) \neq \text{span}(S_1) \cap \text{span}(S_2)$ (15)

indeed

$\text{span}(S_1 \cap S_2) \subset \text{span}(S_1) \cap \text{span}(S_2)$

in this case.

b)

Let

$S_1 = \left\{ \begin{bmatrix} 0 \\ a \end{bmatrix} \mid a \in \mathbb{N} \right\} = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \ldots \right\}$ (16)

$S_2 = \left\{ \begin{bmatrix} 0 \\ b \end{bmatrix} \mid b \in \mathbb{N}^{2k} \right\} = \left\{ \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \end{bmatrix}, \ldots \right\}$ (17)

$S_1 \cap S_2 = S_2$ (18) $\Rightarrow$

$\text{span}(S_1 \cap S_2) = \text{span}(S_2)$ (19)

Also,

$\text{span}(S_1) \cap \text{span}(S_2) = \text{span}(S_2)$ (20)

Actually, $\text{span}(S_1) = \text{span}(S_2)$ in this case, since S_1 & S_2 sets are the same line points (vectors).

Considering (19) & (20):

$$\text{span}(S_1 \cap S_2) = \text{span}(S_1) \cap \text{span}(S_2) \quad (21)$$

which we aimed to show.

$\square$